AI for SMEs — Handbook

An evidence brief based on the selected sources, with a separately labelled implementation proposal.

Start here

This edition distinguishes published findings from suggested next steps. It does not present a fictional SME as a researched case. A source about product design, a research prototype or aggregate statistics is not evidence of inventory savings at a particular company.

The questions this handbook answers

- What do the selected sources actually establish?
- What remains unknown before applying those findings to an SME?
- How could a team investigate an application without assuming benefits?

Evidence from the selected sources

S1 — Hyperspectral Imaging Textiles (HIT)

Selected passage from the supplied English translation:

> The research project HIT develops automatic recognition of textile composition. Mixtures of fibers and existing additives make sorting and high-quality recycling difficult. Existing clothing labels do not always provide sufficient reliable information.

S1

Selected passage from the supplied English translation:

> The team combines hyperspectral SWIR-cameras with AI. Reference samples are investigated; images and information about the composition form data for recognition models. The project works on a dataset, algorithms, and a prototype for classification of textiles on a conveyor belt.

S1

S2 — Het onwerkelijke laten ontstaan: hoe Nike de meest verrassende Air schoenen ooit maakte

Selected passage from the supplied English translation:

> Nike describes A.I.R. as a design project involving thirteen athletes. The participating sports are athletics, football, basketball, and tennis. Participating design teams used information about the athletes' needs to develop visual ideas for sports shoes using generative AI.

S2

Selected passage from the supplied English translation:

> The images served as inspiration, not as direct, usable end products. Designers selected concepts, processed feedback, and investigated practical applications. In further development, they used more digital simulations and three-dimensional prints. Human design knowledge was necessary to recognize impractical presentations.

S2

S3 — Marktsegment Retail - Retail management

Selected passage from the supplied English translation:

> SBB explains how AI can help with retail management. By using past sales data, we can guess what sales might be in the future. This helps with decisions about how much stock to keep and how much to produce.

S3

Selected passage from the supplied English translation:

> The article also mentions product recommendations, customer inquiries, product data, and personnel scheduling as areas where AI is used. The data needed, however, depends on the problem being solved. For supply chain applications, besides purchases, data from suppliers and logistics might also be important.

S3

Selected passage from the supplied English translation:

> The article discusses retail in general. It does not measure the effectiveness of AI for a sports clothing company, nor does it provide a fixed percentage of savings, a budget, or a guaranteed payback period. For a sports apparel application, a separate evaluation of your own data and business processes is still required. This is a limitation of the application, not a reported research result.

S3

Suggested implementation approach — not a reported case

The steps below are educational proposals, not measured results from the sources. The availability of company data, feasibility, budget and expected return must be established locally.

Illustrative use case (CRISP-DM)

- **Business understanding:** Select one concrete decision and define a measurable objective with the business owner. Do not assume a numerical benefit.
- **Data understanding:** Check which records exist, who owns them, and whether they represent the chosen problem. Do not assume an ERP system or historical dataset is available.
- **Data preparation:** Assess missing values, units, permissions and sensitive fields before choosing preparation steps.
- **Modelling:** Compare a simple baseline with a candidate method suitable for the available data; AI is not automatically the preferred option.
- **Evaluation:** Test on held-out observations and assess errors and business consequences against the baseline.
- **Deployment:** Start with a supervised pilot, record decisions and monitor performance before expanding.

Tooling and effort

Decision	Proposed approach	Effort and cost
Data preparation	Assess existing spreadsheet/database tools before buying new software	Not established by this brief; obtain a scoped estimate
Evaluation	Use a reproducible baseline and documented comparison	Not established by this brief; measure during a pilot

Responsible AI and trust

Proposal: review data access, privacy, source attribution and human oversight. Anonymisation or a passing automated check alone does not establish legal compliance or factual accuracy.

First five steps for an owner

1. Choose one business decision to investigate.
2. Check what evidence and permitted data are available.
3. Establish a baseline and agree on evaluation criteria.
4. Run a limited, supervised comparison.
5. Review the results before deciding on further deployment.

Regional ecosystem and support

No regional service or funding recommendation is inferred automatically from the selected material. Verify any proposed provider and its current services separately.

Sources

S1: Hyperspectral Imaging Textiles (HIT)

- Publisher: NHL Stenden Hogeschool
- Publication date: niet vastgesteld op de geraadpleegde pagina
- Access date supplied with the source: 25 oktober 2026
- Supplied material: beknopte AI-samenvatting in eigen woorden; geen volledige artikeltekst.
- Translation file: hyperspectral_imaging_textiles_hit_txt_b06d968e0f1b.txt
- Original and translation provenance were retained separately; the quotation above is from the English translation, not the original publisher wording.
- Original URL: https://www.nhlstenden.com/projecten/hyperspectral-imaging-textiles-hit

S2: Het onwerkelijke laten ontstaan: hoe Nike de meest verrassende Air schoenen ooit maakte

- Publisher: NIKE, Inc.
- Publication date: 11 april 2024
- Access date supplied with the source: 25 oktober 2026
- Supplied material: beknopte AI-samenvatting in eigen woorden; geen volledige artikeltekst.
- Translation file: nikex_air_shoes_innovation_summary_txt_00c58c684f3b.txt
- Original and translation provenance were retained separately; the quotation above is from the English translation, not the original publisher wording.
- Original URL: https://about.nike.com/nl/magazine/creating-the-unreal-how-nike-made-its-wildest-air-footwear-yet

S3: Marktsegment Retail - Retail management

- Publisher: SBB
- Publication date: niet vastgesteld op de geraadpleegde pagina
- Access date supplied with the source: 25 oktober 2026
- Supplied material: beknopte AI-samenvatting van relevante onderdelen; geen volledige artikeltekst.
- Translation file: retail_management_summary_txt_4f63e5185665.txt
- Original and translation provenance were retained separately; the quotation above is from the English translation, not the original publisher wording.
- Original URL: https://trendrapportage.s-bb.nl/trendverkenningen/kunstmatige-intelligentie/marktsegment-retail/?item=retail+management&tab=per+dossier

About this handbook

Prepared for the AIBS × AEL educational project. Research findings are linked to supplied sources; implementation proposals are labelled separately. Human review is required before publication. This document is educational material, not consultancy or a guarantee of outcomes.